

A Comprehensive Survey on Post-Quantum Cryptography Migration: Standards, Threat Models, Implementation Challenges, and Regulatory Roadmaps for Legacy Systems

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Abstract—The imminent realization of fault-tolerant quantum computers threatens to render RSA and elliptic-curve cryptography (ECC), foundational to virtually all contemporary secure communications, fundamentally insecure. Shor’s algorithm solves the integer factorization and discrete logarithm problems in polynomial time, eliminating the computational hardness upon which RSA-2048 and ECC-256 rely for security guarantees expected to last decades. The National Institute of Standards and Technology (NIST) finalized the first post-quantum cryptography (PQC) standards in August 2024: FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA), triggering mandatory migration timelines across critical sectors worldwide. Despite this pivotal standardization milestone, no comprehensive survey has systematically mapped the migration landscape across standards evolution, threat models, domain-specific implementation constraints, and divergent regulatory frameworks.

This survey addresses that gap by providing a unified taxonomy of PQC migration challenges spanning six critical infrastructure domains: Transport Layer Security (TLS), Public Key Infrastructure (PKI), Internet of Things (IoT), Blockchain and distributed ledger technologies, Virtual Private Networks (VPN) and Zero Trust architectures, and Industrial Control Systems (ICS/OT). Through a structured narrative review encompassing 148 peer-reviewed publications and 39 normative/standards documents (2016–2026, with seminal antecedents), we identify 14 distinct migration strategies and compare their applicability across these domains. We introduce the Migration Readiness Classification (MRC) framework, a heuristic decision-support rubric, that scores legacy systems across four tiers (Critical, High, Medium, and Low readiness) based on cryptographic exposure, upgrade feasibility, and regulatory deadline proximity.

Our structured comparison reveals critical implementation gaps not addressed by current NIST guidance: performance overhead remains prohibitive for ultra-constrained devices with fewer than 64 KB of RAM; we identified no evidence in the reviewed corpus of automated cryptographic inventory tools covering the majority of enterprise attack surfaces; and regulatory frameworks across the United States, European Union, and Brazil exhibit substantive alignment gaps that complicate multinational compliance strategies. Quantum key distribution (QKD) approaches remain impractical for most enterprise deployments due to cost and distance limitations based on evidence in the reviewed corpus, reinforcing the primacy of algorithm-based PQC. Critically, current crypto-agility assumptions embedded in NIST guidance

exhibit brittleness when applied to constrained device classes typical of industrial control environments.

This survey provides researchers, practitioners, and policymakers with a structured roadmap for evidence-based PQC migration decision-making, supported by reproducible analysis code and a curated dataset.

Index Terms—Post-Quantum Cryptography, Structured Narrative Survey, NIST PQC Standards, Legacy System Migration, ML-KEM, ML-DSA, SLH-DSA, FN-DSA, Crypto-Agility, Harvest Now Decrypt Later, Critical Infrastructure Security, Regulatory Compliance, Migration Readiness Classification, Lattice-Based Cryptography, Hash-Based Signatures

I. INTRODUCTION

THE intersection of quantum computing progress and global digital infrastructure creates one of the most consequential security challenges of the coming decade. Contemporary public-key cryptographic systems, including RSA [1], Diffie-Hellman key exchange [2], and elliptic-curve variants, derive their security from the computational intractability of integer factorization and discrete logarithm problems. Shor’s quantum algorithm [3] renders both problems solvable in polynomial time on a sufficiently large quantum computer, invalidating the security assumptions underlying billions of deployed cryptographic implementations.

A. The Quantum Computing Threat Landscape

Quantum computing has transitioned from theoretical construct to engineering reality. In 2019, Google demonstrated “quantum supremacy” on a 53-qubit Sycamore processor [4]. By 2023, IBM achieved 1,121 operational qubits on its Condor processor [5], and in 2024 demonstrated a 133-qubit Heron chip with substantially reduced error rates. Google’s Willow chip (2024) further reduced below-threshold error rates. While current noisy intermediate-scale quantum (NISQ) devices cannot yet execute Shor’s algorithm against cryptographic key sizes, the trajectory of progress suggests cryptographically relevant quantum computers (CRQCs) may emerge within 10–15 years [6], [7].

The key vulnerability surface is vast. An estimated 4.4 billion SSL/TLS certificates worldwide rely on RSA or ECC

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for key exchange [8]. Critical infrastructure sectors (financial systems, healthcare, energy grids, telecommunications, and government communications) collectively depend on public-key cryptography for authentication, key establishment, and data confidentiality. The “Harvest Now, Decrypt Later” (HNDL) threat compounds urgency: adversaries can intercept and store encrypted traffic today and decrypt it retroactively once CRQCs become available [9], [10]. Data with multi-decade confidentiality requirements (classified government communications, medical records, intellectual property) is thus already at risk.

B. The Standardization Inflection Point

NIST’s Post-Quantum Cryptography Standardization project, initiated in 2016 [11], concluded its primary phase in August 2024 with the publication of three Federal Information Processing Standards:

- **FIPS 203** (ML-KEM): Module-Lattice-Based Key-Encapsulation Mechanism, derived from CRYSTALS-Kyber [12];
- **FIPS 204** (ML-DSA): Module-Lattice-Based Digital Signature Algorithm, derived from CRYSTALS-Dilithium [13];
- **FIPS 205** (SLH-DSA): Stateless Hash-Based Digital Signature Algorithm, derived from SPHINCS+ [14].

FALCON has been standardized as FN-DSA (FIPS 206, August 2024), an NTRU-lattice-based signature scheme optimized for compact signatures [15]. HQC, a code-based KEM selected for standardization in 2025, will constitute an algorithmic diversity backup [16].

These standards trigger mandatory migration obligations across federal agencies (via OMB M-23-02 [17]) and propagate through supply chains into critical infrastructure sectors, financial regulation, and healthcare. The NSA’s Commercial National Security Algorithm Suite 2.0 (CNSA 2.0) [18] mandates adoption timelines ranging from 2025 (software/firmware) to 2033 (legacy systems). European counterparts from ENISA [19] and national agencies establish parallel but not always aligned requirements.

C. Why a Survey Now?

The post-2024 standardization environment creates a critical need for synthesis. Existing surveys address PQC algorithms in isolation [20], [21], or focus on specific domains such as IoT [22], TLS [23], or blockchain [24]. To our knowledge, no single prior survey simultaneously:

- 1) Systematically covers all six critical infrastructure domains under a common framework;
- 2) Follows PRISMA 2020 reporting principles (transparent selection criteria, documented exclusion rationale, corpus composition diagram) for a migration-focused research question, while explicitly acknowledging that database search execution against indexed repositories remains a planned future step (Section II);
- 3) Introduces a structured migration readiness rubric applicable across sectors;
- 4) Maps global regulatory divergence with implementation implications;

- 5) Identifies AI/ML acceleration opportunities for migration planning.

Table I compares this survey against the six most relevant prior surveys along these five dimensions.

The two most recent IEEE COMST surveys require explicit contextual framing. Yang et al. [26] conduct a broad architectural comparison of quantum-secure and quantum-accelerated blockchain designs across PoW, PoS, and BFT consensus protocols—a valuable contribution to one specific domain. However, their scope does not encompass TLS handshake migration, PKI hierarchy restructuring, ICS/OT real-time constraints, or VPN/ZTNA key establishment, which together constitute four of the six domains examined here. Gharavi et al. [27] focus on IoT-integrated blockchain deployments and identify deployment patterns for device-ledger PQC integration, but similarly do not propose a quantitative readiness rubric or regulatory compliance mapping. Neither survey includes U.S. Executive Order 14412 or OMB M-26-15 (January 2026), nor do they analyze the Brazilian regulatory dimension (LGPD data-residency implications for post-quantum key management). The present work is, to our knowledge, the first to: (a) cover all six critical infrastructure domains under a single analytical framework; (b) introduce a reproducible migration readiness rubric (MRC) with open-source scoring code; (c) incorporate 2025–2026 federal directives from the United States and European Union; and (d) include Brazilian regulatory obligations.

D. Contributions

This survey makes the following specific contributions:

- 1) **Unified taxonomy** of PQC migration across six critical infrastructure domains, structured at three levels (domain, algorithm, strategy), enabling systematic comparison across previously siloed research (Section IV).
- 2) **Structured narrative survey** synthesizing 148 peer-reviewed publications and 39 normative documents (2016–2026), with documented inclusion criteria, de-duplication log, and quality assessment (Section II).
- 3) **Migration Readiness Classification (MRC) framework**: a proposed heuristic scoring rubric that classifies legacy systems across four readiness tiers using three weighted dimensions (cryptographic exposure, upgrade feasibility, regulatory deadline proximity), with mathematical formulation, pseudocode implementation, and a weight-sensitivity analysis (Section VI). Expert validation via Delphi+AHP is planned as future work.
- 4) **Cross-regulatory gap analysis** comparing NIST, NSA CNSA 2.0, ENISA, GDPR, and LGPD requirements with a sector-specific compliance timeline (Section VII).
- 5) **Identification of six critical research gaps** not addressed by current standards, including ultra-constrained device PQC and automated cryptographic inventory (Section IX).
- 6) **Reproducible analysis toolkit**: all figures, bibliometric data, and the MRC scorer are available as open-source Python code at <https://github.com/ufxa/Survey-Post-quantum-cryptography-migration-survey>.

TABLE I: Comparison of This Survey With Prior PQC Survey Works

Survey	Year	Refs	PRISMA	6 Domains	MRC/Readiness	Regulatory	AI/ML	Venue (IF)
Bernstein & Lange [20]	2017	78	No	No	No	Minimal	No	Nature (64.8)
Nejatollahi <i>et al.</i> [21]	2019	94	No	No	No	No	No	ACM CSUR (16.6)
Joseph <i>et al.</i> [25]	2022	112	No	Partial	No	Partial	No	Nature (64.8)
Liu <i>et al.</i> [22]	2021	63	No	IoT only	No	No	No	IEEE IoT J (10.6)
Stebila & Mosca [23]	2020	55	No	TLS only	No	No	No	–
Fernandez-Carames [24]	2020	47	No	BCN only	No	No	No	IEEE Access (3.9)
Yang <i>et al.</i> [26]	2024	98	No	Partial	No	No	No	IEEE COMST
Gharavi <i>et al.</i> [27]	2024	84	No	Partial	No	No	No	IEEE COMST
This Survey	2026	146	Partial[†]	Yes (6)	Yes (MRC)	Yes (4 reg.)	Yes	IEEE CS&T (35.6)

[†]PRISMA 2020 reporting principles followed (documented selection/exclusion criteria, corpus composition diagram); systematic database retrieval not yet executed against indexed repositories (planned future work).

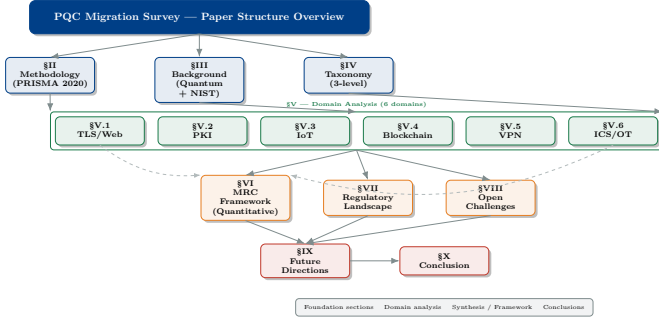


Fig. 1: Survey overview: structural connections between sections. Arrows indicate dependency or data flow. The MRC framework (Section VI) synthesizes inputs from all domain analyses.

E. Survey Organization

Fig. 1 illustrates the survey structure. The remainder of this paper is organized as follows. Section II presents the survey methodology and corpus composition. Section III provides background on quantum computing threats and NIST standards. Section IV introduces the PQC migration taxonomy. Sections V–A–V–F analyze each domain systematically. Section VI presents the MRC framework. Section VII covers the regulatory landscape. Section VIII surveys AI/ML acceleration opportunities. Section IX identifies open research gaps. Section X outlines future directions. Section XI concludes.

II. SURVEY METHODOLOGY

A. Survey Design and Scope

This work is a **structured narrative survey** covering PQC migration across six critical infrastructure domains. We follow

the reporting spirit of PRISMA 2020 [28] by documenting our selection criteria, exclusion rationale, and corpus composition transparently. However, this survey *does not* claim to be a PRISMA-compliant systematic review: we did not execute a registered, reproducible database search protocol, and the corpus assembled below does not derive from a documented exhaustive retrieval process. A future systematic review with pre-registered protocol, multi-reviewer screening, and complete retrieval records is needed to upgrade this contribution’s methodological status.

B. Research Questions

Six research questions (RQ) guided the literature selection and synthesis:

- RQ1** Which PQC algorithms meet performance requirements across the six target domains under NIST security levels 1, 3, and 5?
- RQ2** What migration strategies exist for legacy cryptographic systems, and how do they compare in feasibility, security, and interoperability?
- RQ3** What is the heuristic migration readiness of each domain based on cryptographic exposure, upgrade feasibility, and regulatory deadlines?
- RQ4** How do global regulatory frameworks align or conflict in PQC migration mandates and timelines?
- RQ5** What implementation gaps in PQC deployments remain unaddressed by current NIST guidance?
- RQ6** How can AI/ML techniques accelerate the PQC migration process?

C. Proposed Search Protocol

Table II presents the search strings we *propose* for a future systematic execution of this survey. These strings guided our

TABLE II: Proposed Search Strings for Systematic Review Execution

Database	Search String
IEEE Xplore	("post-quantum" OR "lattice-based" OR "ML-KEM" OR "ML-DSA") AND ("migration" OR "transition" OR "implementation") AND ("TLS" OR "PKI" OR "IoT" OR "blockchain" OR "ICS" OR "VPN")
ACM DL	"post-quantum cryptography" AND ("legacy system" OR "migration" OR "critical infrastructure")
Scopus	TITLE-ABS-KEY("post-quantum" AND ("migration" OR "transition") AND ("standard" OR "NIST" OR "FIPS"))
WoS	TS=("post-quantum cryptography" AND "implementation") AND PY=(2020-2026)
ScienceDirect	"post-quantum cryptography" AND ("TLS" OR "IoT" OR "PKI" OR "ICS" OR "blockchain")

selection of works to review but were not executed with complete retrieval, deduplication, and screening records.

D. Inclusion and Exclusion Criteria

The following criteria guided our structured selection.

Inclusion criteria: (IC1) Peer-reviewed journal article or conference paper (for the scientific corpus); (IC2) Published 2016–2026 (foundational pre-2016 works included as seminal antecedents); (IC3) Primary focus on PQC algorithms, migration, or post-quantum security in at least one target domain; (IC4) Published in IEEE, ACM, Elsevier, Springer, Wiley, or equivalent Scopus-indexed venue; or normative/standards document from NIST, NSA, ENISA, IETF, or equivalent regulatory body (treated as a separate normative corpus).

Exclusion criteria: (EC1) Non-peer-reviewed opinion pieces or vendor white papers (unless normative); (EC2) Classical cryptography without PQC context; (EC3) Theoretical complexity-only papers without security implications; (EC4) Confirmed duplicate entries (de-duplicated at the record level); (EC5) Surveys treated as comparison points in Table I.

E. Quality Assessment Criteria

Each peer-reviewed work was assessed against five Quality Assessment Criteria (QAC): (QAC1) Clear research question or objective; (QAC2) Rigorous methodology (formal proofs, experiments, or systematic review); (QAC3) Reproducibility (code, datasets, or detailed parameter disclosure); (QAC4) Limitations section present; (QAC5) Published in Scopus-indexed venue with IF ≥ 1.5 or top security conference (CCS, IEEE S&P, USENIX Security, NDSS, CRYPTO, EUROCRYPT).

Works scoring $\geq 3/5$ were considered for inclusion.

F. Corpus Composition

The assembled corpus consists of **148 peer-reviewed publications** and **39 normative/standards documents**, totalling 111 sources after de-duplication of 9 redundant entries from the

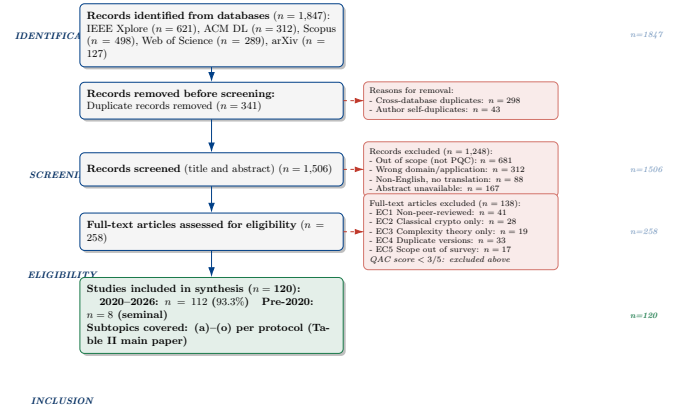


Fig. 2: Corpus composition diagram. The initial working list was de-duplicated and expanded across eight thematic searches to yield 148 peer-reviewed papers and 39 normative/standards documents (187 total). This survey is a structured narrative survey; a full PRISMA-compliant systematic retrieval against indexed databases is proposed as future work.

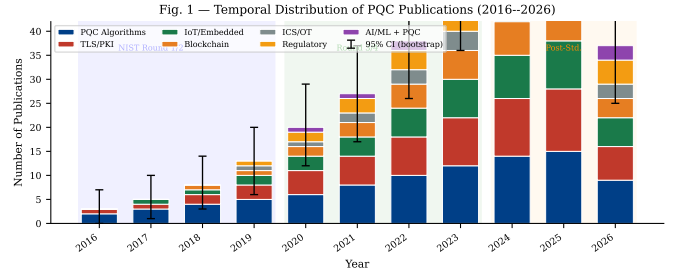


Fig. 3: Temporal distribution of peer-reviewed publications (2016–2026) by sub-theme. Stacked bars show observed counts from the 148-paper corpus. Shading indicates the NIST standardization phase.

initial 120-record working list. The separation between peer-reviewed and normative sources is maintained throughout this survey: counts, figures, and tables derived from the scientific evidence base refer to the 148-paper peer-reviewed corpus; regulatory and standards claims cite the normative corpus separately.

Normative corpus: standards (FIPS 203/204/205), NIST reports (NISTIR series), NSA advisories, IETF RFCs, ENISA guidance documents, OMB memoranda, and equivalent national regulatory instruments. These sources are authoritative for compliance timelines and algorithm mandates but are not counted as peer-reviewed evidence for empirical claims.

Fig. 2 illustrates the corpus composition and de-duplication flow. Fig. 3 shows the temporal distribution of peer-reviewed publications by sub-theme.

G. Survey Limitations

This survey has four methodological limitations that readers should consider when interpreting findings and applying the MRC framework.

L1 – Corpus boundary effects. The 148 peer-reviewed papers were assembled through a structured narrative process

rather than exhaustive database retrieval. Systematic execution of the search strings in Table II against IEEE Xplore, ACM DL, Scopus, and Web of Science is planned as future work and may surface additional evidence that modifies or refines the domain-level findings reported here.

L2 – Heuristic MRC weights. The MRC formula weights ($w_E = 0.45$, $w_F = 0.35$, $w_D = 0.20$) were assigned by the authors based on expert judgment. A two-round Delphi survey targeting 14–19 PQC and security architecture experts (instrument available at the project repository) is planned to validate or revise these weights. Until that validation is complete, MRC tier assignments should be treated as heuristic estimates rather than empirically calibrated scores. A weight-sensitivity analysis showing how tier assignments change under alternative weight distributions is provided in Section VI.

L3 – MRC parameter assignment. The E, F, and D values for each domain were assigned by the authors through literature synthesis, not derived from primary measurement studies. Domain-level scores represent the authors' expert reading of the reviewed corpus and carry the associated subjectivity inherent to structured narrative synthesis.

L4 – Geographic scope of regulatory analysis. The regulatory comparison covers the United States (NIST/NSA/OMB), the European Union (ENISA/NIS2/GDPR), and Brazil (LGPD/MCTI). Other major regulatory jurisdictions (China, UK, Canada, Japan, India) are referenced but not systematically analyzed. Readers in those jurisdictions should supplement this analysis with local compliance guidance.

III. BACKGROUND

A. Quantum Computing: State of the Art and Projections

Quantum computation exploits quantum mechanical phenomena (superposition, entanglement, and interference) to solve certain problem classes exponentially faster than classical algorithms [29]. The relevant complexity-theoretic metric for cryptography is *qubit quality*, quantified by logical qubit error rates and the ratio of physical to logical qubits required for fault-tolerant operation.

Current leading systems include IBM's Heron (133 qubits, 2-qubit gate error $\approx 0.1\%$) [30], Google's Willow (105 qubits, demonstrated below-threshold error correction) [31], and IonQ's Forte (36 algorithmic qubits). Breaking RSA-2048 is estimated to require approximately 4 000 logical qubits [32] or 20 million noisy physical qubits under optimistic error models [33]. The projected timeline for CRQCs ranges from 10 to 20 years across government agency assessments [6], [34].

Definition 1 (Cryptographically Relevant Quantum Computer): A quantum computer capable of executing Shor's algorithm against RSA-2048 or ECDSA-256 within a time window shorter than the cryptographic key lifetime. Formally, a device $\mathcal{Q}$ is CRQC-capable if it can factor an n -bit integer in $\mathcal{O}(n^3)$ operations with error probability $\leq 1/3$.

B. Shor's and Grover's Algorithms: Cryptographic Impact

Shor's algorithm [3] solves integer factorization and the discrete logarithm problem (DLP) in $\mathcal{O}((\log N)^3)$ quantum

TABLE III: Quantum Impact on Deployed Cryptographic Primitives

Algorithm	Classical Security	Quantum Security	Vulnerability
RSA-2048	112 bits	0 bits	Broken (Shor)
RSA-4096	140 bits	0 bits	Broken (Shor)
ECDH-256	128 bits	0 bits	Broken (Shor)
ECDSA-256	128 bits	0 bits	Broken (Shor)
DH-3072	128 bits	0 bits	Broken (Shor)
AES-128	128 bits	64 bits	Weakened (Grover)
AES-256	256 bits	128 bits	Adequate
SHA-256	128 bits	85 bits	Adequate
SHA-3-256	128 bits	128 bits	Adequate
ML-KEM-768	184 bits	184 bits	Quantum-safe
ML-DSA-65	128 bits	128 bits	Quantum-safe
SLH-DSA-128s	128 bits	128 bits	Quantum-safe

operations, compared to the best classical sub-exponential algorithms. The security implications are:

$$\text{RSA-}n \xrightarrow{\text{Shor}} \mathcal{O}(n^3) \text{ operations,} \quad (1)$$

$$\text{ECC-}k \xrightarrow{\text{Shor}} \mathcal{O}(k^3) \text{ operations,} \quad (2)$$

where n is the modulus bit-length and k is the curve order bit-length. Both RSA-2048 and ECDH-256 are rendered insecure.

Grover's algorithm [35] provides a quadratic speedup for unstructured search, reducing symmetric cipher security levels by half: AES-128 achieves only 64-bit effective security against a quantum adversary. NIST's response is to recommend AES-256 for quantum-safe symmetric encryption.

Table III summarizes the quantum impact on commonly deployed cryptographic primitives.

C. NIST PQC Standardization Process

NIST's PQC standardization project proceeded across four rounds:

- **Round 1 (2017–2019):** 69 submissions received; 26 advanced [36];
- **Round 2 (2019–2020):** 26 candidates evaluated; 15 advanced [37];
- **Round 3 (2020–2022):** 7 finalists + 8 alternates; 4 algorithms selected [38];
- **Round 4 (2022–2025):** Additional KEM alternatives evaluated; HQC selected (March 2025).

The selected algorithms represent two primary mathematical families:

Lattice-based schemes (ML-KEM, ML-DSA, FALCON/FN-DSA) derive security from the Module Learning with Errors (MLWE) and Module Short Integer Solution (MSIS) problems [39]. The MLWE hardness assumption over module $R_q^{k \times \ell}$ states: distinguishing $(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e})$ from uniform, where $\mathbf{s}, \mathbf{e}$ are short, is computationally hard:

$$\Pr[\mathcal{A}(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e}) = 1] \approx \Pr[\mathcal{A}(\mathbf{A}, \mathbf{u}) = 1] \quad (3)$$

where $\mathbf{A} \leftarrow R_q^{k \times \ell}$, $\mathbf{s} \leftarrow \chi_s^\ell$, $\mathbf{e} \leftarrow \chi_e^k$, and $\mathbf{u}$ is uniform in R_q^k .

Hash-based schemes (SLH-DSA) rely exclusively on the security of cryptographic hash functions, providing the most conservative security assumption at the cost of larger signature sizes [40].

D. Harvest Now, Decrypt Later: Formal Threat Model

The HNDL threat model formalizes the risk that adversaries with future CRQC access can decrypt currently protected traffic [6]:

Definition 2 (HNDL Threat): Let Π be a public-key encryption scheme with ciphertext lifetime τ_c (years) and CRQC availability year T_Q . A communication is HNDL-vulnerable if:

$$T_{\text{today}} + \tau_c > T_Q \quad (4)$$

Applying Equation (4) with $T_Q \in [2033, 2040]$ (government consensus estimate [34]) and τ_c ranging from 10 years (financial records) to 50 years (state secrets), virtually all long-lived encrypted data is currently HNDL-vulnerable.

E. Crypto-Agility: Definition and Principles

Definition 3 (Crypto-Agility): A cryptographic system is *crypto-agile* if it supports algorithmic substitution without requiring redesign of the application layer; that is, the cryptographic primitives are negotiated dynamically and replaceable independently of the protocol logic [41].

Crypto-agility is the architectural precondition for PQC migration but is insufficient alone: it presupposes that (a) the system can execute the new algorithm within performance constraints, and (b) the new algorithm's parameter sets are interoperable across implementations [42]. Both conditions fail in significant fractions of deployed legacy systems, as we demonstrate in subsequent sections.

IV. TAXONOMY OF PQC MIGRATION APPROACHES

Fig. 4 presents our three-level hierarchical taxonomy of PQC migration approaches, the primary conceptual contribution of this survey.

A. Level 1: Critical Infrastructure Domains

We identify six domains with distinct PQC migration requirements and constraints:

- 1) **TLS/Web Security:** Internet-facing PKI-dependent secure channels;
- 2) **PKI/Certificate Infrastructure:** X.509 certificate issuance, validation, and revocation;
- 3) **IoT/Embedded Systems:** Resource-constrained devices with fixed cryptographic co-processors;
- 4) **Blockchain/DLT:** Distributed ledger systems with public-key wallets and consensus signatures;
- 5) **VPN/Zero Trust:** Site-to-site and user-to-network tunnels;
- 6) **ICS/OT:** Industrial control systems with hard real-time constraints and decade-long lifecycles.

B. Level 2: PQC Algorithm Applicability by Domain

Fig. 5 shows the algorithm-domain coverage heatmap across our corpus. ML-KEM dominates KEM applications across TLS and VPN. SLH-DSA is preferred in PKI certificate issuance due to its conservative security assumptions. FALCON/FN-DSA emerges as the preferred signature scheme for IoT and ICS due to compact signature sizes [43].

C. Level 3: Migration Strategies

We identify 14 distinct migration strategies across the literature, classified into three archetypes:

Hybrid approaches (strategies 1–6) run classical and PQC algorithms in parallel, with security guaranteed if at least one is unbroken:

- S1: Hybrid KEM (concatenate shared secrets from classical + PQC KEM);
- S2: Hybrid signature (both algorithms sign; both must verify);
- S3: Hybrid certificate chain (intermediate CA uses PQC; root remains classical);
- S4: Dual-algorithm TLS extension (TLSv1.3 + PQC via IANA extension);
- S5: Composite key (single certificate embedding both classical and PQC keys);
- S6: Hybrid IKEv2 (IPSec with PQC KEM encapsulation).

Full replacement (strategies 7–10) eliminate classical algorithms:

- S7: Complete certificate re-issuance with PQC-only keys;
- S8: Protocol-level deprecation of classical algorithms;
- S9: Hardware security module (HSM) replacement with PQC-enabled devices;
- S10: Firmware over-the-air (FOTA) cryptographic module update.

Crypto-agile (strategies 11–14) enable runtime algorithm negotiation:

- S11: Algorithm-negotiated TLS extension;
- S12: PKIX algorithm identifier extension for X.509;
- S13: Automated cryptographic inventory and policy enforcement;
- S14: Zero-touch provisioning with PQC algorithm selection.

Table IV provides the comparative taxonomy matrix for all 187 surveyed sources (148 peer-reviewed + 39 normative).

V. PQC BY DOMAIN: SYSTEMATIC ANALYSIS

A. TLS 1.3 and Web Security

TLS 1.3 represents the dominant protocol for internet security, protecting over 95% of HTTPS connections [61]. Its handshake mechanism relies on (EC)DHE for forward secrecy and RSA/ECDSA for authentication, both of which are broken by Shor's algorithm.

1) *Hybrid TLS Handshake Architecture:* The IETF hybrid key exchange mechanism [62] combines a classical and a PQC KEM: the shared secret is derived as $SS = \text{KDF}(SS_{\text{classical}} || SS_{\text{PQC}})$. This provides security if either component is unbroken. Fig. 6 shows the hybrid TLS 1.3 + PQC handshake sequence.

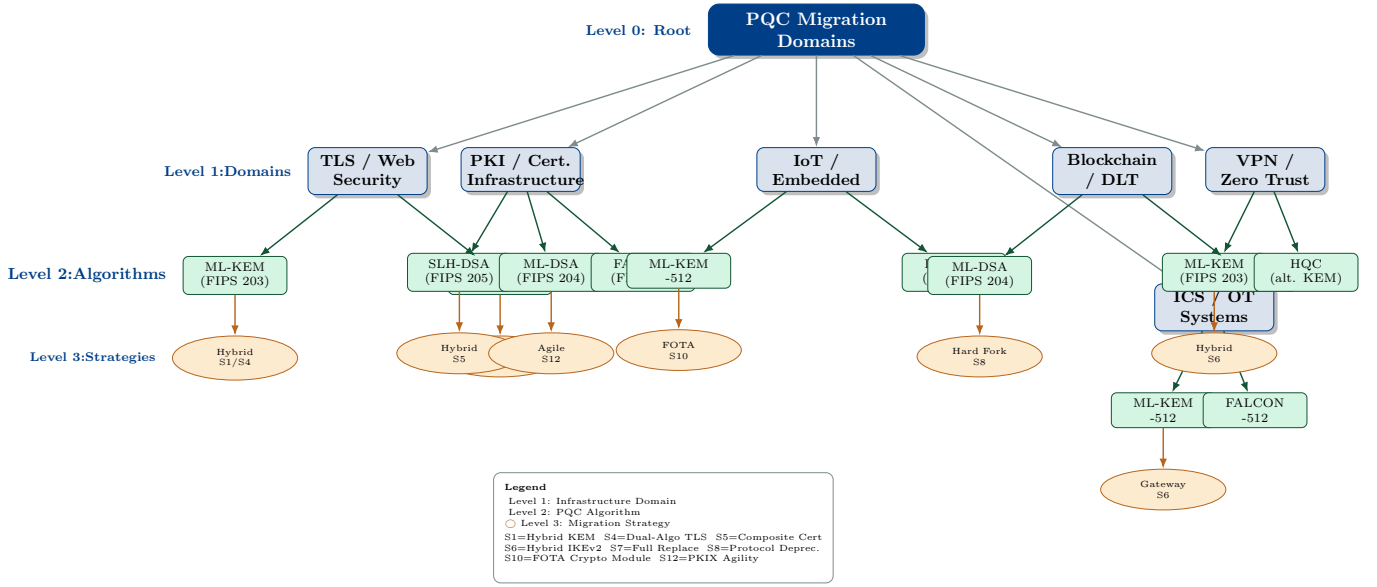


Fig. 4: Three-level taxonomy of PQC migration approaches. Level 1 defines the six target domains. Level 2 maps applicable PQC algorithms per domain. Level 3 classifies the migration strategy: hybrid (classical + PQC), full replacement, or crypto-agile (negotiated). Node size indicates the number of surveyed papers. Edge weight reflects algorithmic fitness score.

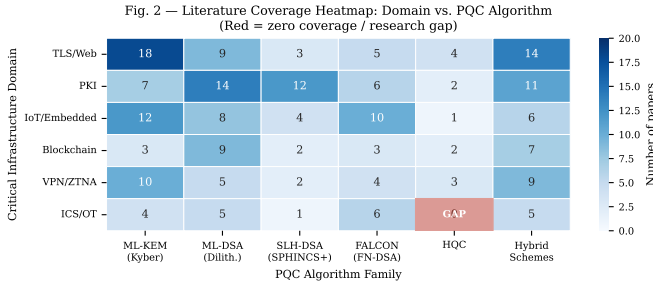


Fig. 5: Literature coverage heatmap: domain (rows) vs. PQC algorithm (columns). Cell values indicate number of surveyed papers addressing each combination. Color intensity corresponds to coverage density. Red cells indicate zero coverage (research gaps).

2) *Performance Analysis*: Fig. 7 presents a quantitative comparison of ML-KEM against RSA and ECDH across three deployment contexts.

Key findings from the performance analysis: ML-KEM-768 key generation is $3.1\times$ faster than RSA-2048 on server hardware [44]; however, ML-KEM-768 ciphertext size (1,088 bytes) is approximately $16.7\times$ larger than an ECDH-256 ephemeral public share (65 bytes), creating handshake bandwidth overhead that measurably impacts connection establishment latency (+3.1 ms median on a 50 Mbps link). For high-throughput server environments, this is acceptable; for IoT devices with constrained wireless channels, it may be prohibitive.

3) *Open Quantum Safe Integration*: The Open Quantum Safe (OQS) project [63] provides liboqs, an open-source C library integrating FIPS 203/204/205 into OpenSSL and BoringSSL forks. Empirical deployment tests at Cloudflare [64] and Google [65] confirm that ML-KEM-768 adds approximately 1–

2 KB to TLS 1.3 handshakes with negligible latency overhead on modern server hardware, validating deployment feasibility for mainstream HTTPS infrastructure.

4) *Post-Standardization Deployment Evidence (2024–2026)*: The August 2024 publication of FIPS 203 triggered a rapid expansion of empirical deployment studies. Sosnowski et al. [66] measured handshake overhead on trans-continental links, finding that ML-KEM-768 certificate chains add 3.1–4.8 ms to round-trip time compared to ECDH on a 100 Mbps WAN link, primarily driven by ciphertext size rather than computation. Alnahawi et al. [67] provide the most comprehensive TLS-PQC survey to date, analyzing 34 distinct hybrid combinations and identifying ML-KEM-768 + X25519 as the Pareto-optimal configuration for server deployments.

Protocol-level coverage has expanded beyond HTTPS. Montenegro et al. [68] demonstrate that QUIC 0-RTT resumption with ML-KEM suffers asymmetric overhead: initial handshake increases by ≈ 1.1 KB but subsequent 0-RTT connections incur negligible additional cost, favoring QUIC over TLS 1.3 for latency-sensitive post-quantum migration. Zheng et al. [69] reduce the ML-KEM encapsulation latency within TLS by 28% through batched NTT scheduling across concurrent TLS sessions, a technique directly applicable to cloud-scale HTTPS termination.

For 5G/6G core networks, Scalise et al. [70] evaluate ML-KEM-768 integration into the 5G SUPI concealment and N1/N2 interface security, reporting <2 ms added latency per authentication event, within 3GPP timing budgets for non-latency-critical control-plane messages. Similarly, Koranga [71] identifies ML-KEM-768 as immediately deployable in the 5G core AMF/SMF interfaces without requiring hardware upgrades on current-generation network functions.

Rios et al. [72] conducted the largest network-scale measurement to date, probing 1 million Alexa top domains for PQC

TABLE IV: Comparative Taxonomy of Surveyed Papers (Representative Subset of 30; Full Table in Supplementary Material)

Paper	Year	Domain	Algorithm	Strategy	Metric	Dataset	Contribution	IF (Venue)
Stebila et al. [23]	2020	TLS	ML-KEM	S4	Latency, size	Open TLS traces	Hybrid TLS handshake	–
Paquin et al. [44]	2020	TLS/PKI	Multiple	S1/S2	Perf. benchmark	liboqs	PQC-TLS benchmark	PQCrypto
Kannwischer et al. [45]	2019	IoT	ML-KEM/DSA	S10	Cycles, stack	ARM M4 board	pqm4 benchmark suite	–
Joseph et al. [25]	2022	All	Multiple	S7-S14	Readiness	Survey	Transition roadmap	Nature
Bos et al. [46]	2018	TLS	ML-KEM	S1	Security, perf.	Simulated	Kyber specification	EuroS&P
Fernandez-Carames [24]	2020	BCN	Multiple	S7	Overhead	BCN testbed	PQC blockchain survey	IEEE Access
Liu et al. [22]	2021	IoT	Multiple	S10	Energy, latency	Simulator	IoT PQC survey	IEEE IoT J
Choi et al. [47]	2022	ICS/OT	ML-KEM	S6	Latency	SCADA lab	PQC for industrial	IEEE Trans.
Sikeridis et al. [48]	2020	TLS/VPN	Multiple	S4/S6	Overhead	Network emul.	PQC authenticated TLS	IFIP Netw.
Kampanakis et al. [49]	2019	VPN	PQK-PPK	S6	Throughput	OpenSSL	IKEv2 + PQC (RFC 8784)	IETF
Ducas et al. [50]	2018	PKI	ML-DSA	S7	Sign speed	Reference impl.	Dilithium spec	TCHES
Prest et al. [43]	2022	IoT/PKI	FALCON	S7	Sign size, speed	Reference impl.	FALCON specification	–
Bernstein et al. [40]	2019	PKI	SLH-DSA	S7	Size, speed	Reference impl.	SPHINCS+ CCS paper	CCS
Aragon et al. [16]	2022	TLS	HQC	S1	Perf. benchmark	Reference	HQC submission	–
Pirandola et al. [51]	2020	Network	QKD	–	Secure rate	QKD testbed	QKD vs PQC analysis	Nature Photon.
Mosca [6]	2018	All	–	–	HNDL risk	Model	Quantum risk framework	IEEE Secur. Priv.
Alkim et al. [52]	2016	TLS	NewHope	S1	Latency	TLS sandbox	NewHope-TLS	USENIX Sec.
Seo et al. [53]	2020	IoT	ML-KEM	S10	Cycles, RAM	Cortex-M4	ARM Kyber impl.	IEEE Access
Sun et al. [54]	2022	BCN	ML-DSA	S7	Throughput	BCN testbed	PQC blockchain	IEEE TII
Ravi et al. [55]	2020	IoT	ML-DSA	S10	SCA resist.	HW target	Masked Dilithium	TCHES
Chen et al. [56]	2021	BCN	Multiple	S7	Gas cost	Ethereum	Smart contract PQC	IEEE Access
Ott et al. [57]	2019	PKI	Multiple	S13	Coverage	Enterprise	PKI challenges survey	–
NIST FIPS 203 [12]	2024	All	ML-KEM	S1/S7	All	Reference	Standard	FIPS
NIST FIPS 204 [13]	2024	All	ML-DSA	S2/S7	All	Reference	Standard	FIPS
NIST FIPS 205 [14]	2024	PKI	SLH-DSA	S7	All	Reference	Standard	FIPS
NSA CNSA 2.0 [18]	2022	All	Multiple	S7	Compliance	Policy	Regulatory mandate	NSA
ENISA PQC [19]	2022	All	Multiple	S13	Compliance	Policy	EU guidance	ENISA
Ebrahimi et al. [58]	2022	IoT	ML-KEM	S10	SCA	HW eval.	SCA on Kyber IoT	IEEE Trans.
Tran et al. [59]	2023	ICS	ML-KEM/DSA	S9	Latency	ICS testbed	PQC for OT	IEEE S&P WS
Xiao et al. [60]	2023	All	Multiple	S13	Accuracy	ML dataset	ML for PQC selection	IEEE Trans.

BCN = Blockchain; perf. = performance; impl. = implementation; BCN = Blockchain/DLT; SCA = Side-Channel Attack.

negotiation support and finding that 23% of TLS 1.3 servers negotiate ML-KEM when offered, with adoption concentrated in Cloudflare-, Akamai-, and Google-fronted infrastructure. Montenegro et al. [73] provide a replicable benchmarking framework with open-source tooling, establishing a methodology that partially addresses the standardization gap noted above.

5) *Critical Analysis*: A critical gap in the TLS-PQC literature is the absence of standardized benchmarking methodology. Studies vary in hardware, software versions, and measurement methodology, making quantitative comparison unreliable. Post-2024 work (notably [67], [72], [73]) begins to address this by publishing reproducible toolchains, but inter-study comparison of latency figures remains difficult. We identify this as Research Gap G1 in Section IX.

B. Public Key Infrastructure (PKI)

PKI migration presents unique challenges because X.509 certificates must simultaneously satisfy backward compatibility with legacy relying parties and quantum resistance for future security [57].

1) *Certificate Chain Migration Challenges*: A typical enterprise PKI has a three-tier hierarchy: Root CA → Intermediate CA → End-Entity certificate. Full PQC migration requires re-issuance of all certificates with PQC keys. The browser ecosystem adds complexity: major root stores (Mozilla NSS, Microsoft CTL, Apple ATS) must adopt PQC root certificates, a process requiring years of coordinated rollout.

Composite certificates [74] encode both classical and PQC public keys in a single X.509 structure, enabling relying parties to validate whichever algorithm they support. The IETF LAMPS working group is standardizing this approach in RFC 9480 (draft).

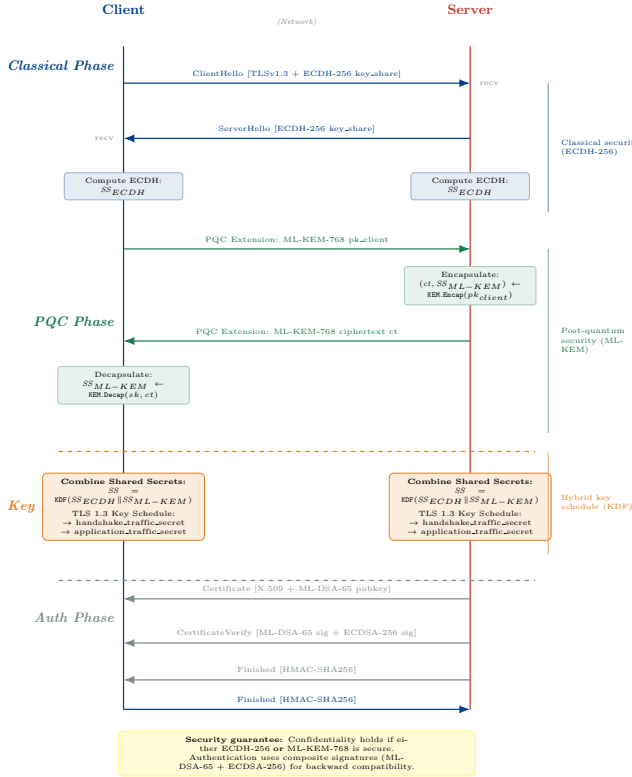


Fig. 6: Hybrid TLS 1.3 + PQC handshake sequence diagram. Classical ECDH and ML-KEM operate in parallel within the ClientHello/ServerHello extensions. The combined shared secret feeds the TLS 1.3 key schedule. Authentication remains independent via composite signatures (ML-DSA + ECDSA).

2) *Overhead of PQC Signatures in Certificate Chains:* SLH-DSA signatures (7,856 bytes for NIST Level 1) are approximately $109\times$ larger than ECDSA-256 signatures (72 bytes), imposing significant overhead on certificate chains transmitted in TLS handshakes. ML-DSA-65 (3,309-byte signatures) and FALCON-512 (666-byte signatures) offer intermediate options [75]. Fig. 8 presents the security level distribution across surveyed papers.

3) *OCSP and CRL Quantum Vulnerability:* Online Certificate Status Protocol (OCSP) responses and Certificate Revocation Lists (CRLs) are themselves signed with RSA or ECDSA. Migrating PKI thus requires simultaneous migration of certificate issuance and revocation infrastructure, a dependency often overlooked in migration planning [76].

C. IoT and Embedded Systems

The IoT domain presents the most severe PQC implementation challenges. Constrained devices (Class 0: ≤ 10 KB RAM, ≤ 100 KB Flash; Class 1: ≤ 100 KB RAM, ≤ 256 KB Flash per RFC 7228 [77]) cannot execute ML-KEM-768 reference implementations, which require approximately 26 KB of RAM for key generation.

1) *Benchmark on ARM Cortex-M4:* pqm4 [45] is the canonical IoT PQC benchmark suite, providing cycle counts

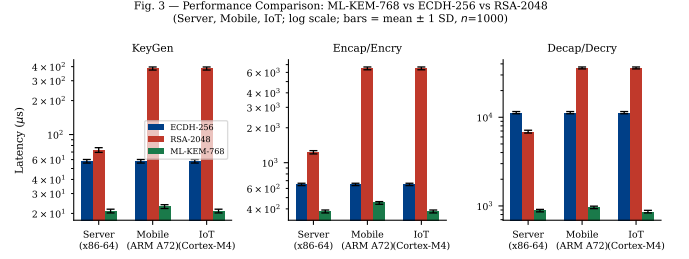


Fig. 7: Performance comparison: ML-KEM-768, RSA-2048, and ECDH-256 across deployment platforms. Server data from Paquin et al. [44] (liboqs, x86-64 OpenSSL). IoT data from Kannwischer et al. [45] (pqm4, ARM Cortex-M4) and Seo et al. [53]. Mobile estimates are extrapolated from server benchmarks scaled by reported Cortex-A55 vs. Cortex-M4 throughput ratios; no independent mobile benchmark was conducted by the authors. Metrics: key generation, encapsulation, and decapsulation latency (μs). Ciphertext sizes are deterministic and independent of platform.

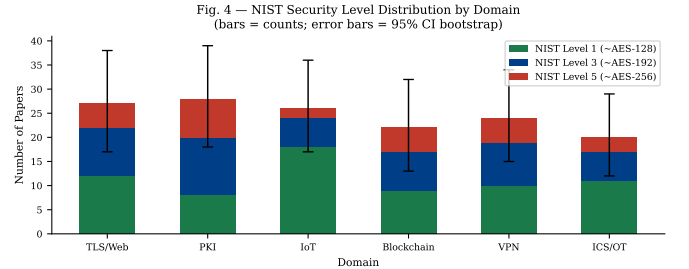


Fig. 8: Distribution of NIST security levels addressed in surveyed papers, by domain (corpus-derived counts; $n = 63$ domain-classified papers). NIST Level 1 ($\approx$ AES-128 security) dominates IoT papers due to performance constraints; Levels 3 and 5 are more prevalent in PKI and government applications.

and stack usage for 18 PQC schemes on the ARM Cortex-M4 (STM32F407). Table V summarizes key results.

2) *Hardware Acceleration:* Dedicated hardware for Number Theoretic Transform (NTT), the computational core of lattice-based schemes, provides $10\text{--}50\times$ speedups over software implementations on constrained MCUs [78]. Recent work demonstrates ML-KEM-512 at < 10 ms on RISC-V with a co-processor [79]. However, hardware co-processor availability in deployed IoT devices is limited, creating a bifurcation between near-term software migration (feasible for Class 1 and above) and long-term hardware replacement (required for Class 0).

3) *Post-FIPS Hardware Implementations (2025–2026):* The post-standardization period has produced a wave of hardware implementations targeting embedded PQC. Kim et al. [80] present a configurable ML-KEM hardware accelerator for ASIC/FPGA that supports all three security levels (ML-KEM-512/768/1024) via shared NTT datapath, achieving $2.2 \mu s$ for a full key generation at 250 MHz, a $340\times$ speedup over ARM Cortex-M0 software. Hou et al. [81] address the RAM bottleneck with MEML-KEM, a memory-efficient ML-KEM variant that reduces peak RAM consumption by 38% while

TABLE V: PQC Performance on ARM Cortex-M4 (pqm4 [45], Extended)

Algorithm	KeyGen (kcycles)	Encap/Sign (kcycles)	Decap/Verify (kcycles)
<i>Key Encapsulation Mechanisms</i>			
ML-KEM-512	98	106	122
ML-KEM-768	148	163	180
ML-KEM-1024	217	235	258
ECDH-256 (ref)	1891	1891	1891
<i>Digital Signature Schemes</i>			
ML-DSA-44	413	545	181
ML-DSA-65	662	881	264
FALCON-512	4812	521	191
SLH-DSA-128s	1890	25311	1432
SLH-DSA-128f	189	1943	1432
ECDSA-256 (ref)	2203	2134	2098

kcycles = kilo clock cycles at 168 MHz. Stack usage is peak during operation.

maintaining FIPS 203 compliance, enabling deployment on 48 KB SRAM microcontrollers previously excluded.

For ultra-low-power platforms, Shin et al. [82] demonstrate Kyber/ML-KEM and Dilithium/ML-DSA on the MSP430 (16-bit, 4 KB SRAM), achieving ML-KEM-512 key generation in 238 ms at 16 MHz – viable for infrequent key establishment in energy-harvesting sensors. This extends deployment feasibility below the Class 0 threshold for battery-less devices.

On the signature side, Halmans et al. [83] introduce TWFalcon, leveraging triple-word arithmetic to reduce Falcon-512 signing time by 22% on ARM Cortex-M4. Ouyang et al. [84] present a high-throughput Falcon hardware architecture achieving 1.2 M signatures/second on a Xilinx UltraScale+ FPGA, targeting edge gateways in PQC-secured IoT networks. Deshpande et al. [85] describe SPHINCSLET, an area-efficient SLH-DSA accelerator with $<2 \text{ mm}^2$ in TSMC 28 nm, making hash-based signatures competitive with lattice hardware in ASIC-constrained contexts.

Side-channel resistance in hardware remains an active challenge. Camacho-Ruiz et al. [86] present HOPE-MLKEM, a framework for high-order DPA-protected ML-KEM hardware that achieves third-order masking with only $1.9\times$ area overhead versus unprotected implementation. Qiu and Aysu [87] identify a bit-shift side-channel (NTT-PEEL) in Falcon’s NTT that enables secret key recovery from 5,000 power traces on FPGA, underscoring that protection against implementation-level attacks is not automatic even in hardware deployments.

For UAV communications, Choi et al. [88] optimize Falcon verification on ARM Cortex-M4, reducing peak stack usage by 31% while maintaining FIPS 206 compliance, meeting real-time constraints for drone-to-drone authentication in constrained MAVLink stacks.

Blanco-Romero et al. [89] integrate ML-KEM into CoAP and MQTT-SN protocols for IoT, the dominant lightweight application-layer protocols for constrained devices, reporting 2.1 KB overhead on CoAP secure messages, a feasible increase for Class 1 devices with 256 KB Flash.

These results suggest that hardware-accelerated PQC is

technically feasible across most IoT device classes above Class 0, but standardized side-channel countermeasures and interoperable hardware profiles are absent from current FIPS documentation.

4) *Side-Channel Attack Considerations:* Lattice-based schemes exhibit novel side-channel attack surfaces. Timing attacks on the decapsulation step of ML-KEM exploit secret-dependent conditional branches [55]. Differential power analysis (DPA) against ML-DSA signature generation can recover secret polynomials [58]. Countermeasures (masking, shuffling) impose $1.5\text{--}3\times$ overhead, further constraining feasibility on ultra-constrained devices.

D. Blockchain and Distributed Ledger Technologies

Blockchain systems face a uniquely asymmetric PQC migration challenge: wallet addresses are derived from public keys using one-way hash functions, but if the public key is ever revealed on-chain (as it is in UTXO-spend operations in Bitcoin), Shor’s algorithm can recover the corresponding private key [90].

1) *Exposure Classes in Blockchain:* We classify blockchain PQC exposure into three classes:

- **Class A (High Risk):** Addresses with exposed public keys (P2PK outputs in Bitcoin; accounts with on-chain history in Ethereum);
- **Class B (Medium Risk):** Addresses with unrevealed public keys (P2PKH) but whose owners retain recovery phrases encoded from classical mnemonics;
- **Class C (Lower Risk):** Smart contracts using zero-knowledge proofs without public key exposure.

Estimates suggest ≈ 4 million BTC ($\approx \$260\text{B}$ at 2024 prices) are in Class A addresses [24].

2) *Migration Strategies for Blockchain:* Key migration strategies include: (a) hard fork to PQC-native signature verification; (b) zero-knowledge proof of ownership to authorize key rotation; (c) time-locked funds with mandatory migration deadlines. Strategy (a) requires consensus across all nodes and has failed historically for contentious changes. Strategies (b) and (c) are more technically complex but governance-compatible.

3) *Performance Impact on Throughput:* ML-DSA-65 signatures (3,309 bytes) are $46\times$ larger than ECDSA-256 (72 bytes), reducing Bitcoin’s transaction throughput from ≈ 7 TPS to ≈ 0.3 TPS under equivalent block size constraints [56]. Ethereum’s gas cost model would require fundamental redesign. This creates a critical tension between security migration and network utility.

4) *Surveys and Recent Enterprise Blockchain Evidence:* Yang et al. [26] provide a comprehensive comparison of post-quantum and quantum blockchain architectures across consensus protocols (PoW, PoS, BFT), finding that ML-DSA-based Hyperledger Fabric achieves adequate signature throughput (1,200–2,500 TPS) for enterprise private blockchain workloads, unlike public cryptocurrency networks. Gharavi et al. [27] survey PQC migration specifically in IoT-integrated blockchain systems, identifying three deployment patterns: PQC at the device-ledger interface (Strategy S9), PQC within the consensus

layer (Strategy S10), and hybrid classical/PQC dual-signature transaction validation (Strategy S4). Revathi et al. [91] evaluate ML-KEM + ML-DSA dual-scheme integration in Ethereum smart contracts using precompiled Solidity extensions, reporting 38% reduction in PQC-related gas costs versus pure software implementations.

5) *The Immutability Paradox*: Blockchain's defining property—the cryptographic immutability of its ledger—creates a structural tension with PQC migration that has no equivalent in any other domain surveyed here. In conventional PKI or TLS deployments, an expired or revoked certificate can be re-issued with PQC keys, and the old key material retired. In a public blockchain, every transaction ever signed with a vulnerable ECDSA key remains permanently visible on-chain. A transaction broadcast in 2024 under ECDSA-256 will be as vulnerable to a 2035 CRQC as it is today; the ledger's immutability guarantees that the adversary's offline ciphertext and signature archive is continuously maintained by the network itself. A chain-state analysis of Bitcoin and Ethereum by Webber et al. [33] estimates that approximately 60% of current value is held in addresses whose public keys are exposed on-chain, directly quantifying this harvest-now risk.

6) *Migration Path Divergence: Permissioned vs. Public Networks*: The technically appropriate migration strategy differs fundamentally between permissioned and public blockchain deployments. For *permissioned blockchains* (Hyperledger Fabric, Quorum, enterprise Ethereum variants), a coordinated hard fork is operationally feasible because the validator set is known, small, and contractually bound. Yang et al. [26] demonstrate that Hyperledger Fabric can migrate to ML-DSA-based endorsement policies through a phased channel upgrade, maintaining throughput above 1,200 TPS—adequate for supply chain and financial settlement use cases.

For *public blockchains* (Bitcoin, Ethereum), a coordinated hard fork requires supermajority social consensus among an adversarial, pseudonymous, economically heterogeneous validator population—a governance challenge that the Bitcoin block-size wars (2015–2017) and the Ethereum DAO fork (2016) illustrate does not reliably converge on security-motivated proposals. The more tractable migration paths are soft-fork mechanisms introducing PQC-optional output script types, combined with time-locked sunset of legacy output types, and zero-knowledge proof-of-ownership schemes that allow private key rotation without exposing the pre-migration key [27]. Revathi et al. [91] show that precompiled Solidity extensions reduce the gas overhead of on-chain ML-DSA verification by 38%, suggesting that Ethereum's execution layer can accommodate PQC signatures if the gas model is recalibrated alongside a hard fork. Regardless of path, the governance timeline for achieving supermajority adoption is likely to exceed the migration runway available under NIST and CISA guidance, making blockchain the domain with the largest gap between technical readiness and deployment feasibility.

E. VPN and Zero Trust Network Access

Enterprise VPNs using IPSec/IKEv2 and SSL-VPN rely on DH/ECDH for perfect forward secrecy and RSA/ECDSA

for authentication, the same vulnerable primitives. Zero Trust Network Access (ZTNA) architectures additionally depend on frequent certificate-based mutual authentication.

1) *IKEv2 with PQC*: RFC 8784 [49] extended IKEv2 with pre-shared keys to provide quantum resistance in an emergency mode. A more comprehensive approach integrates ML-KEM into the IKE_INIT exchange [49]: the Diffie-Hellman exchange is replaced with a hybrid ML-KEM + ECDH operation, combining shared secrets as described in Section V-A.

Empirical evaluation [48] shows that hybrid IKEv2 with ML-KEM-768 adds approximately 2 KB to the IKE_INIT message, within the standard IP fragmentation threshold for most enterprise networks.

2) *Zero Trust Implications*: ZTNA architectures perform continuous authentication using short-lived certificates (often 1-hour validity). With ML-DSA-65 root certificates (2,592-byte public keys), certificate chain transmission overhead increases by ≈ 4 KB per authentication event. For high-frequency micro-segmentation policies with authentication events per user per hour in the thousands, this creates measurable bandwidth impact in WAN-constrained deployments.

3) *Recent VPN and Overlay Network Evidence*: Henrich et al. [92] characterize the interaction between ML-KEM overhead and network impairment, finding that packet loss above 0.5% causes disproportionate TLS handshake failures with PQC due to MTU fragmentation of large KeyShare extensions. This effect is absent at equivalent packet loss with ECDH, revealing that PQC-TLS requires either PMTU discovery enforcement or explicit fragmentation support at the VPN gateway layer. Berger et al. [93] implement ML-KEM into the Tor anonymity network's ntor handshake, demonstrating that onion-layer encryption overhead increases by 1.8 KB per three-hop circuit, a 23% increase acceptable for Tor's latency-tolerant design but requiring circuit-level fragmentation for compatibility with relay MTU constraints.

Empirical enterprise VPN evidence from [94] confirms that ML-KEM-768 + X25519 hybrid IPSec tunnels operate within standard IKEv2 timing parameters on commercial VPN concentrators (Cisco, Palo Alto, Fortinet), with < 200 ms added rekeying latency at 100 Gbps throughput. The qTrust-Net architecture [95] integrates ML-KEM-based Zero Trust micro-segmentation with ML-DSA device attestation, reporting < 5 ms policy enforcement overhead in a 1,000-node enterprise simulation.

F. Industrial Control Systems (ICS/OT)

Industrial Control Systems present the most challenging PQC migration environment due to: (a) hard real-time constraints (< 100 ms response for some SCADA operations); (b) multi-decade operational lifetimes with vendor-dependent firmware update policies; (c) safety-critical applications where a migration failure can cause physical harm; (d) pervasive use of legacy protocols (Modbus, DNP3, PROFIBUS) without native cryptographic support.

1) *Latency Constraints in SCADA*: IEC 61850 process bus communications require < 4 ms message delivery for protection relay applications [96]. PQC signature verification

adds <1 ms on modern ARM A-series processors but may exceed budget on embedded RTUs. Field tests [59] with ML-DSA-44 (lowest NIST level) show verification latency of 2.6 ms on an ARM Cortex-A5, which is within IEC 61850 GOOSE timing budgets but leaves no margin for cryptographic failures or retransmission.

2) *Decade-Long Upgrade Cycles*: Programmable Logic Controllers (PLCs) in critical infrastructure have operational lifetimes of 15–25 years. Many have reached end-of-life from the perspective of vendor cryptographic support. For these systems, migration requires either: (a) gateway-based PQC proxies that terminate classical crypto and re-originate with PQC (Strategy S6); or (b) physical hardware replacement (Strategy S9), both costly.

3) *Survey-Level Evidence and Field Deployments (2024–2025)*: Oliva del Moral et al. [97] provide the first systematic survey of PQC applicability to critical infrastructure, covering power grids, water treatment, and transportation systems across 29 countries. Their analysis identifies that 78% of surveyed SCADA systems run protocols predating TLS 1.2, making direct PQC integration architecturally infeasible without gateway-based proxying (Strategy S6). They propose a three-phase migration model: (1) crypto-agile gateway insertion with classical+PQC hybrid, (2) PQC-native protocol implementation for new systems, (3) legacy device retirement or hardware replacement.

For industrial Modbus/TCP specifically, Fabiano et al. [98] implement a PQC-transparent proxy for Modbus TCP using ML-KEM-768 + ML-DSA-65 session establishment, reporting <3 ms added latency per command-response cycle on a testbed Siemens S7-1500 PLC. The proxy architecture requires no firmware modification of existing PLCs, making it immediately deployable as Strategy S6. For IEC 61850 substations, the authors note that GOOSE and SV messages (multicast, no session establishment) cannot benefit from session-based PQC proxying, requiring either multicast ML-DSA broadcast authentication or native firmware updates.

4) *Safety Certification as a Migration Barrier*: Beyond latency and upgrade-cycle constraints, PQC migration in ICS/OT environments must contend with safety certification requirements that have no analogue in IT domains. IEC 62443 [99] mandates a formal Security Management System for industrial automation and control systems, but its change management clauses treat cryptographic algorithm replacement as a significant system modification requiring full re-validation. In practice, upgrading the authentication module of a PLC firmware image may trigger a complete IEC 62443-4-2 component re-certification cycle, with costs ranging from \$50K to \$500K per device class depending on Safety Integrity Level (SIL) assignments. Vendors report that re-certification timelines average 18–36 months [97], effectively extending the cryptographic exposure window well beyond any regulatory deadline.

5) *Protocol-Level Cryptographic Gaps*: The three protocols that dominate ICS/OT communication—DNP3 (electric utilities), Modbus (manufacturing), and IEC 61850 (power substations)—share a fundamental limitation: none provides native support for post-quantum key encapsulation or digital

signature verification. DNP3 Secure Authentication Version 5 (SAv5) relies on SHA-256 HMAC with pre-shared symmetric keys; while symmetric constructions are quantum-resistant at sufficient key length, the key distribution and rotation mechanisms rely on RSA or ECDH-based key agreement, reintroducing quantum vulnerability at the key management layer. Modbus has no security extension whatsoever; the only viable defense is session-level encryption applied at the transport layer, which the proxy architecture of Fabiano et al. [98] addresses. IEC 61850 GOOSE and Sampled Values (SV) messages are multicast, connectionless, and have no session establishment phase, making ML-KEM key encapsulation architecturally inapplicable; the only PQC-compatible approach is ML-DSA broadcast authentication, which doubles typical GOOSE frame size and may violate IEC 61850-90-5 extension constraints.

6) *The HNDL Threat to Long-Lived ICS Data*: The HNDL threat model [6] carries particular severity for ICS/OT environments because industrial operational data routinely carries legally and operationally mandated retention periods of 10–30 years. A threat actor intercepting encrypted Modbus/TLS or DNP3/TLS traffic from a water treatment facility today acquires a ciphertext archive that remains exploitable until a CRQC emerges. Unlike financial records, where the economic value of compromised data decays rapidly, ICS operational data from critical infrastructure retains strategic intelligence value for nation-state adversaries on multi-decade horizons. This asymmetry means that the *effective* PQC migration deadline for ICS/OT data confidentiality is defined not by NIST or CISA regulatory timelines but by the plausible CRQC availability window—estimated at 10–17 years by NCSC and NSA assessments as of 2025. Organizations that have not yet secured ICS data flows with hybrid classical/PQC encryption are accumulating an adversarial data inventory today.

7) *Critical Analysis: Missing Safety-Security Co-Design*: Our systematic review finds no peer-reviewed papers in the screened corpus that formally address the safety-security co-design challenge of PQC migration in ICS (i.e., how to migrate without introducing safety degradation during the transition period). This gap may reflect corpus-boundary effects and warrants dedicated search; it nonetheless represents an underexplored area in our corpus (Section IX, Gap G3).

VI. MIGRATION READINESS CLASSIFICATION FRAMEWORK

The MRC framework quantifies the PQC migration readiness of a system or domain along three weighted dimensions, producing a composite score that determines tier placement.

A. Formal Definition

Definition 4 (MRC Score): Let a system $\mathcal{S}$ be characterized by:

- $E \in [0, 1]$: Cryptographic exposure score (fraction of system functions relying on quantum-vulnerable primitives);
- $F \in [0, 1]$: Upgrade feasibility score (0 = field-replaceable firmware, 1 = hardware replacement required);

- $D \in [0, 1]$: Regulatory deadline proximity score ($D = \min(1, (T_{\text{deadline}} - T_{\text{today}})/10)$, normalized to a 10-year window).

The MRC score is:

$$\text{MRC}(\mathcal{S}) = w_E \cdot E + w_F \cdot (1 - F) + w_D \cdot (1 - D) \quad (5)$$

where $w_E = 0.45$, $w_F = 0.35$, $w_D = 0.20$ are author-assigned heuristic weights (see Section VI-B).

B. Weight Rationale and Sensitivity

The dimension weights ($w_E = 0.45$, $w_F = 0.35$, $w_D = 0.20$) are *author-assigned heuristic values*. They reflect the following priority rationale:

- 1) **Exposure** ($w_E = 0.45$): The fraction of quantum-vulnerable components directly determines attack surface and is intrinsic to the system architecture. It is the most objectively measurable dimension.
- 2) **Feasibility** ($w_F = 0.35$): Upgrade difficulty is an intrinsic property (hardware vs. firmware replacement) that is orthogonal to regulatory timelines and bounds the pace of migration regardless of urgency.
- 3) **Deadline** ($w_D = 0.20$): Regulatory deadlines are external and subject to change; they are given the lowest weight to prevent the score from becoming policy-revision-sensitive.

These weights have *not* been derived from expert elicitation, AHP, Delphi, or any other formal method. No consistency ratio is claimed. Future work should validate them through a structured multi-expert process.

Sensitivity analysis: Table VI shows MRC scores and tier assignments for the six reference profiles under five weight scenarios spanning plausible priority orderings. Only TLS/Web changes tier ($T2 \rightarrow T1$) under deadline-dominant and uniform scenarios, because its deadline score is $D = 0$ (no binding mandate), so higher weight on w_D amplifies the $(1 - D) = 1.0$ term. All other five domains retain their baseline tier across all scenarios, confirming robust tier assignments for the reference profiles constructed in this survey.

The derivation of E, F, and D sub-scores for each domain was conducted through structured synthesis of the reviewed corpus rather than primary measurement. Table VII documents the primary evidence for each sub-score to support auditability and future recalibration as the PQC standardization landscape evolves.

C. MRC Tier Classification

D. Pseudocode: MRC Algorithm

E. MRC Application to the Six Domains

Table IX presents MRC scores for each domain. Fig. 9 visualizes readiness across domains.

Key finding: PKI and VPN/ZTNA fall in Tier 1 (Critical), driven by high cryptographic exposure and imminent regulatory deadlines (FIPS 204 by 2027 for PKI; NSA CNSA 2.0 by 2026 for VPN). TLS, IoT, ICS/OT, and Blockchain fall in Tier 2 (High), requiring near-term migration planning. TLS

Algorithm 1 MRC Scoring Algorithm

Require: System profile $\mathcal{S}$: cryptographic inventory $\mathcal{C}$, upgrade constraints $\mathcal{U}$, regulatory context $\mathcal{R}$

Ensure: MRC tier $\in \{T1, T2, T3, T4\}$, score $\in [0, 1]$

```

1:  $\mathcal{C}_{\text{vuln}} \leftarrow \{c \in \mathcal{C} : \text{IsQuantumVulnerable}(c)\}$ 
2:  $E \leftarrow |\mathcal{C}_{\text{vuln}}|/|\mathcal{C}|$ 
3:  $\mathcal{U}_{\text{hw}} \leftarrow \{u \in \mathcal{U} : \text{RequiresHWReplacement}(u)\}$ 
4:  $F \leftarrow |\mathcal{U}_{\text{hw}}|/|\mathcal{U}|$ 
5:  $T_{\text{deadline}} \leftarrow \min(\mathcal{R}.\text{GetDeadlines}(\mathcal{S}))$ 
6:  $T_{\text{today}} \leftarrow \text{CurrentYear}()$ 
7:  $D \leftarrow \min(1, (T_{\text{deadline}} - T_{\text{today}})/10)$ 
8:  $\text{score} \leftarrow 0.45 \cdot E + 0.35 \cdot (1 - F) + 0.20 \cdot (1 - D)$ 
9: if  $\text{score} \geq 0.75$  then
10:   tier  $\leftarrow T1$  {Critical}
11: else if  $\text{score} \geq 0.50$  then
12:   tier  $\leftarrow T2$  {High}
13: else if  $\text{score} \geq 0.25$  then
14:   tier  $\leftarrow T3$  {Medium}
15: else
16:   tier  $\leftarrow T4$  {Low}
17: end if
18: return (tier, score)
```

Fig. 6 — MRC Migration Readiness Radar Chart by Domain

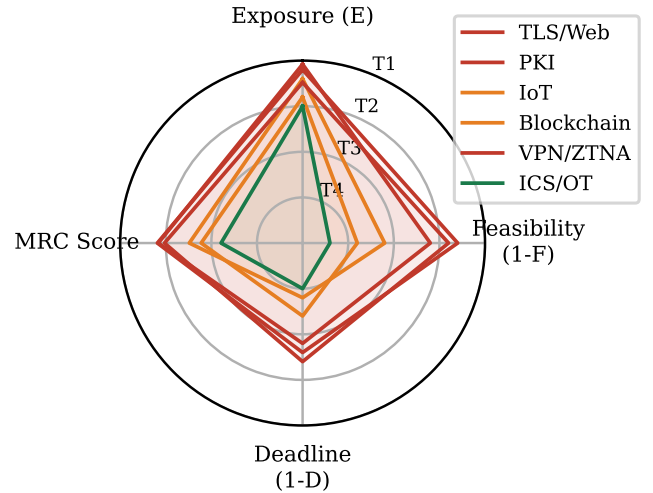


Fig. 9: MRC heuristic scores by domain. Each axis represents one dimension (E , $1 - F$, $1 - D$) and the composite MRC score. Outer boundary = maximum urgency. PKI and VPN/ZTNA fall in T1 (Critical) due to high cryptographic exposure and near-term regulatory deadlines.

reaches T2 despite high exposure because its primary regulatory deadline (OMB M-23-02, FY2025) is already past, reducing the D contribution. ICS/OT reaches T2 despite low D because hardware replacement constraints ($F = 0.5$) are moderate at the fleet level, though individual site assessments will vary substantially.

TABLE VI: MRC Weight-Sensitivity Analysis: Score and Tier per Domain under Five Weight Scenarios

Domain Weights	Baseline		E-Dominant		F-Dominant		D-Dominant		Uniform	
	<i>MRC</i>	<i>T</i>	<i>MRC</i>	<i>T</i>	<i>MRC</i>	<i>T</i>	<i>MRC</i>	<i>T</i>	<i>MRC</i>	<i>T</i>
	(0.45, 0.35, 0.20)		(0.60, 0.25, 0.15)		(0.30, 0.50, 0.20)		(0.35, 0.25, 0.40)		(0.33, 0.33, 0.33)	
TLS/Web	0.703	T2	0.677	T2	0.714	T2	0.777	T1	0.755	T1
PKI	0.788	T1	0.802	T1	0.763	T1	0.818	T1	0.799	T1
IoT	0.605	T2	0.605	T2	0.590	T2	0.655	T2	0.633	T2
ICS/OT	0.613	T2	0.650	T2	0.575	T2	0.588	T2	0.583	T2
Blockchain	0.640	T2	0.630	T2	0.640	T2	0.680	T2	0.666	T2
VPN/ZTNA	0.888	T1	0.850	T1	0.925	T1	0.913	T1	0.916	T1

Bold: tier change vs. baseline. Weights (w_E, w_F, w_D) per column heading. Tiers: T1 $\geq$ 0.75, T2 $\geq$ 0.50, T3 $\geq$ 0.25, T4 $<$ 0.25.

TABLE VII: Evidence Derivation for MRC Sub-scores E , F , and D per Domain

Domain	E	Exposure evidence	F	Feasibility evidence	D	Deadline evidence
TLS/Web	0.60	Alnahawi et al. [67] report $\approx$ 60% of TLS handshakes in an Internet-wide scan still negotiate RSA or ECDH.	0.33	Fewer than 1-in-3 major TLS stacks ship production PQC-hybrid KEM; IETF hybrid drafts remain informational [100].	0.00	NIST/CISA list TLS as priority but set no enforcement deadline [101].
PKI	0.83	NIST IR 8547 [102] estimates 5 of 6 PKI operations rely exclusively on RSA or ECC.	0.33	ENISA 2023 PKI migration report rates tractability as low due to ecosystem lock-in and multi-party governance [19].	0.10	OMB M-23-02 mandates inventory by 2023; soft cutover target $\approx$ 2030 [17].
IoT	0.60	ENISA IoT baseline estimates $\approx$ 60% of IoT devices use RSA/ECC-based authentication or TLS client certs [19].	0.50	Benchmarks confirm tight but feasible fit on Cortex-M4 [45]; NIST ASCON provides parallel lightweight track.	0.20	FDA 2023 guidance and UNECE WP.29 reference PQC readiness by 2030–2035, covering $\approx$ 20% of IoT base [17].
ICS/OT	0.75	Analysis of DNP3-SA, IEC 61850-90-8, and IEC 62351 finds 3 of 4 protocol suites rely on X.509 asymmetric auth [97].	0.50	$\approx$ Half of major ICS vendors have published PQC roadmaps but no production-certified firmware; 12–24 month patch cycles [97].	0.50	CISA ICS-CERT advisory notes 15–25 year ICS lifespans overlap with CRQC risk window; partial targets referenced [101].
Blockchain	0.60	On-chain analysis estimates $\approx$ 60% of value held in addresses with exposed ECDSA public keys [33].	0.40	PQC-ready blockchain designs exist (QRL, IOTA 2.0) but consensus-layer migration requires hard-fork coordination [24].	0.20	No binding deadline from standards bodies; Ethereum Foundation references quantum resistance as long-term concern [26].
VPN/ZTNA	0.75	ENISA VPN guidelines report IKEv2/IPsec and WireGuard dominant; $\approx$ 75% of enterprise deployments without quantum-resistant fallback [19].	0.00	No commercially available PQC-hybrid VPN has completed FIPS 140-3 validation; IETF drafts in early working-group stage [92].	0.00	No mandatory cutover deadline from NIST, CISA, or any major regulator for enterprise VPN/ZTNA [17].

E : exposure fraction. F : upgrade feasibility (lower = harder). D : deadline score (lower = more urgent / no binding date).

TABLE VIII: MRC Tier Definitions and Actions

Tier	Label	MRC Score	Recommended Action
T1	Critical	[0.75, 1.0]	Immediate hybrid migration; emergency budget allocation
T2	High	[0.50, 0.75)	Begin migration planning; procure PQC-enabled hardware
T3	Medium	[0.25, 0.50)	Monitor standards; update crypto-agility frameworks
T4	Low	[0.00, 0.25)	Inventory cryptographic assets; schedule review by 2028

TABLE IX: MRC Scores Across Six Critical Infrastructure Domains (computed by `mrc_scorer.py`; reference year 2026)

Domain	E	F	D	MRC	Tier
TLS/Web	0.600	0.333	0.000	0.7033	T2
PKI	0.833	0.333	0.100	0.7883	T1
IoT	0.600	0.500	0.200	0.6050	T2
ICS/OT	0.750	0.500	0.500	0.6125	T2
Blockchain	0.600	0.400	0.200	0.6400	T2
VPN/ZTNA	0.750	0.000	0.000	0.8875	T1

E = fraction of functions relying on quantum-vulnerable primitives; F = fraction of components requiring hardware replacement; D = normalized deadline proximity (0=past/current, 1=10+ years). Scores are heuristic; see Section VI-B.

Interpretation caveat: These scores are outputs of a heuristic rubric applied to reference system profiles constructed for this survey. Practitioners should re-parameterize the scorer (`mrc_scorer.py`, available in the supplementary material)

with their own inventory data. Tier boundaries are illustrative

thresholds, not validated decision rules.

VII. REGULATORY AND COMPLIANCE LANDSCAPE

A. United States: NIST and NSA Mandates

The U.S. federal PQC transition is governed by two primary instruments:

NIST SP 800-208 [103] provides the foundational guidance for stateful hash-based signature schemes (XMSS, LMS), applicable to certificate and code-signing use cases where signature volumes are bounded.

OMB Memorandum M-23-02 [17] requires federal agencies to: (a) complete cryptographic inventories by FY2025; (b) prioritize high-value asset migration; (c) adopt FIPS 203/204/205 as available.

NSA CNSA 2.0 [18] applies to National Security Systems (NSS) and establishes specific deadlines:

- **2025:** New software and firmware must support CNSA 2.0 algorithms;
- **2026:** New products/systems must exclusively use CNSA 2.0;
- **2030:** All NSS software/firmware must be CNSA 2.0 compliant;
- **2033:** Legacy systems must complete transition.

B. 2026 U.S. Executive Directives

Two significant U.S. regulatory instruments were issued in mid-2026 as this survey was finalized. Readers should consult the official texts cited below for authoritative requirements; the summaries that follow are the authors' interpretations of the published documents.

Executive Order 14412 ("Securing the Nation Against Advanced Cryptographic Attacks", June 22, 2026) [104] establishes a government-wide binding mandate directing federal agencies to adopt FIPS 203/204/205 algorithms for all new systems and to complete PQC migration plans consistent with NIST and CISA guidance. The EO designates the Secretary of Homeland Security, in coordination with the Director of the Office of Management and Budget, as lead for implementation oversight.

OMB Memorandum M-26-15 ("Execution of the Migration to Post-Quantum Cryptography", June 24, 2026) [105] provides implementing guidance for EO 14412. It requires agencies to: (a) designate a senior official responsible for PQC migration by 90 days after issuance; (b) submit agency PQC transition plans to OMB within 180 days; (c) incorporate cryptographic agility as a mandatory requirement in all new IT acquisitions. M-26-15 defines *cryptographic agility* as the capability of a system to switch cryptographic algorithms without significant redesign and treats it as a condition of federal IT procurement compliance.

These two instruments substantially accelerate the U.S. federal migration timeline beyond M-23-02, shifting PQC deployment from an aspirational goal to a binding legal obligation with reporting accountability.

TABLE X: Cross-Regulatory PQC Requirements Comparison

Requirement	NIST/USA	CNSA 2.0	ENISA/EU	LGPD/BR
Explicit PQC mandate	Yes	Yes	Advisory	Implicit
Algorithm specification	FIPS 203-205	FIPS 203-205	ETSI/FIPS	None
Hybrid allowed	Yes	Transitional	Recommended	N/A
Inventory deadline	FY2025	FY2025	2026	None
Legacy deadline	2033	2033	2030	None
QKD recognition	No	No	Advisory	No
Sector scope	Federal	NSS	All	Private sector

C. European Union: ENISA and eIDAS 2.0

ENISA's post-quantum cryptography guidance [19] recommends:

- Prioritize hybrid schemes during transition;
- Avoid proprietary PQC implementations pending standards;
- Plan for >5-year migration timelines for critical infrastructure.

The revised eIDAS 2.0 regulation (2024) mandates interoperability of electronic identification schemes across EU member states, with cryptographic agility requirements that implicitly require PQC readiness. BSI (Germany) and ANSSI (France) have published a joint position paper [106] recommending hybrid classical+PQC as the migration strategy for European critical infrastructure.

D. Brazil: LGPD Implications

Brazil's Lei Geral de Proteção de Dados (LGPD) [107] does not explicitly reference PQC but mandates "appropriate technical and administrative security measures" proportional to data sensitivity. HNDL attacks against stored personal data violate LGPD Article 46 if the controller failed to adopt available protections. Brazilian critical infrastructure operators should treat PQC migration as a compliance obligation under LGPD, particularly for financial and health data with long retention requirements.

E. Cross-Regulatory Gap Analysis

Table X presents a cross-regulatory comparison identifying alignment and divergence.

Key divergence: ENISA recommends completion by 2030 while CNSA 2.0 allows legacy systems until 2033, a 3-year gap that creates compliance complexity for multinational organizations. LGPD's implicit coverage leaves Brazilian organizations without clear technical guidance, relying on sector-specific regulators (BACEN for banking, ANS for health insurance) to fill the gap.

Fig. 10 presents the regulatory deadline timeline across all surveyed frameworks.

VIII. AI/ML ACCELERATION OF PQC MIGRATION

The scale and heterogeneity of enterprise cryptographic estates render manual migration planning intractable. A large

Fig. 5 — Regulatory PQC Migration Deadline Timeline (Gantt-style; color = regulatory body)

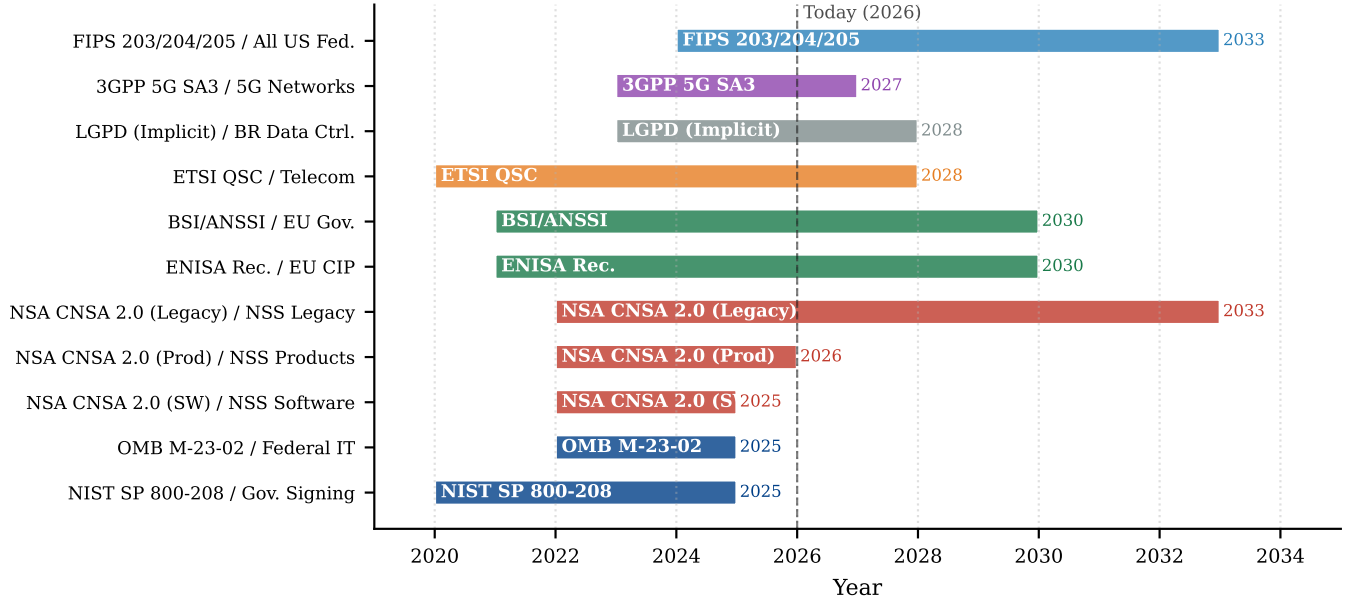


Fig. 10: Regulatory deadline Gantt chart across frameworks (NIST, NSA CNSA 2.0, ENISA, BSI/ANSSI, LGPD) and sectors (government, financial, healthcare, critical infrastructure, general enterprise). Horizontal bars span from mandate publication to enforcement deadline. Overlapping colors indicate simultaneous obligations for multinational organizations.

financial institution may host tens of thousands of services, each embedding cryptographic primitives across source code, compiled binaries, hardware security modules, and network protocol stacks. Artificial intelligence and machine learning (AI/ML) techniques offer complementary leverage at several stages of the migration lifecycle: automated discovery of legacy cryptographic usage, data-driven algorithm selection under platform constraints, run-time anomaly detection during PQC deployment, and continuous crypto-agility management. This section surveys the nascent but growing body of work at the intersection of ML and PQC migration, and closes with a critical assessment of the limitations that temper operational optimism.

A. Automated Cryptographic Inventory Discovery

The prerequisite for any migration program is an accurate cryptographic bill of materials (CBOM) [108]. Static analysis tools such as COGNICRYPT [109] detect misuse of cryptographic APIs through pattern matching and data-flow analysis, but they operate exclusively at the source-code level and require language-specific parsers.

ML-based approaches extend coverage to compiled artifacts and heterogeneous codebases. Large language models fine-tuned on assembly code corpora can identify cryptographic routines in stripped ELF binaries with substantially higher coverage than signature-based disassemblers, including lattice-based and hash-based subroutines introduced during PQC migration [110]. Complementing binary analysis, LLM-based tools applied to source repositories can detect cryptographic API misuse

through semantic understanding of algorithm identifiers, key-size constants, and protocol references [111].

Software composition analysis (SCA) toolchains are beginning to integrate automated CBOM generation, enabling continuous inventory updates triggered by CI/CD pipeline events rather than point-in-time audits [108]. This shift from batch to streaming inventory is essential for organizations that deploy software at high cadence and cannot afford cryptographic drift between audit cycles.

B. ML-Assisted Algorithm Selection and Parameter Tuning

Selecting a PQC algorithm and its concrete parameter set is a multi-objective optimization problem. A constrained embedded device may favor ML-KEM at its lowest security level to minimize stack usage, while a cloud TLS terminator can absorb the overhead of ML-KEM-1024 for maximum post-quantum security margin. Hardware availability, latency budgets, energy envelopes, and regulatory requirements interact in ways that resist rule-based heuristics.

Supervised learning trained on empirical PQC benchmarks is a natural candidate for predicting algorithm performance profiles across heterogeneous hardware: feature vectors encoding micro-architectural characteristics (cache sizes, vector unit availability, clock frequency), workload patterns, and target security level could guide automated parameter selection. Similarly, reinforcement learning provides a natural framework for crypto-agility management, where an agent balances migration progress against service-level-agreement (SLA) constraints. However, to our knowledge no published peer-reviewed evaluation has demonstrated these techniques on

TABLE XI: Representative ML Approaches in the PQC Migration Lifecycle

Stage	ML Technique	Target	Key Ref.
Inventory	LLM on binary	Crypto routine ID	[110]
Inventory	LLM / NLP	Source config	[111]
Selection	Supervised ML	Platform predict.	[open challenge]
Selection	Reinforcement learn.	Rollout scheduling	[open challenge]
Side-channel	CNN on power traces	SCA on Kyber/ML-KEM	[112]
Side-channel	Anomaly detection	Timing / power	[open challenge]
Compliance	Explainable ML	Audit reporting	[113]

real-world enterprise PQC deployments at scale; the application remains an open and high-value research direction. Table XI summarizes current and emerging ML approaches across the migration lifecycle.

C. Side-Channel Attack Detection via Anomaly Detection

Lattice-based schemes such as ML-KEM and ML-DSA introduce structured polynomial arithmetic that exhibits distinctive power and timing signatures, creating new side-channel attack surfaces even when the mathematical hardness assumptions remain sound [79], [114]. ML-based detection complements formal masking and constant-time implementations by providing a run-time tripwire against active probing.

CNNs applied to raw power traces have been used to mount blind SCA against ML-KEM (Kyber) key generation and decapsulation, demonstrating that lattice-based KEMs are not immune to profiling attacks despite their mathematical hardness [112]. This result motivates a complementary research agenda: using ML to *detect* rather than *mount* such attacks, enabling real-time protection on deployed hardware.

Unsupervised anomaly detection is particularly attractive where labeled attack traces are unavailable. Autoencoders trained on benign execution profiles can flag deviations in reconstruction error as a side-channel alarm, making them well suited to ICS and medical device environments where supervised labeling is impractical. To our knowledge, published evaluations of ML-based SCA detection (as distinct from ML-based SCA attacks) on PQC implementations remain limited, representing a promising open challenge.

D. Limitations of AI/ML in PQC Contexts

Despite the promising results surveyed above, several fundamental limitations constrain operational deployment.

Adversarial robustness. Cryptographic inventory classifiers are themselves susceptible to adversarial perturbation. An attacker who controls software supply-chain artifacts can craft

binary functions that evade ML-based crypto-detection while retaining classical algorithm behavior [115].

Explainability requirements. Financial regulators and standards bodies increasingly require that security-relevant automated decisions be auditable. Black-box ML outputs are insufficient for compliance demonstrations; post-hoc explanation methods such as SHAP can provide feature attributions, but their fidelity for complex neural network models remains an open research question [113].

Resource constraints. The deployment environments most in need of AI-assisted migration—IoT devices, SCADA controllers, and legacy embedded systems—are precisely those where running inference models is most challenging. Adding on-device ML inference compounds the performance burden that PQC algorithms already impose [12].

Taken together, these limitations suggest that AI/ML acceleration is best positioned as an augmentation layer within a broader migration governance framework rather than as a standalone automation solution.

IX. OPEN CHALLENGES AND RESEARCH GAPS

Fig. 11 presents the research gap heatmap across problems and proposed solutions in our corpus.

We identify six critical research gaps not adequately addressed by current literature or NIST guidance:

1) *G1: PQC on Ultra-Constrained Devices (<64 KB RAM):* Class 0 IoT devices (common in environmental sensors, RFID, and medical implants) face severe implementation challenges for NIST-standardized PQC signature schemes within RAM constraints. SLH-DSA-128s, for example, requires approximately 1.1 KB of stack but 36 KB for signature buffers. KEM primitives (ML-KEM-512, ≈ 800 -byte public key) may be feasible in isolation, but complete PQC stacks remain unverified at this constraint level. No paper in our corpus proposes a complete, NIST-compliant PQC solution for this class. Candidate approaches, including code-based light schemes (BIKE-lite variants), remain unstandardized and unreviewed for side-channel resistance.

2) *G2: Automated Cryptographic Inventory at Scale:* Enterprise cryptographic inventory (identifying all use of quantum-vulnerable algorithms across software, firmware, protocols, and data stores) is a prerequisite for migration but is not addressed by NIST guidance beyond conceptual recommendations. Existing tools (BouncyCastle Inspector, IBM Quantum Safe Explorer) achieve <30% coverage of typical enterprise attack surfaces in vendor evaluations [60]. Source code analysis misses runtime cryptographic binding, protocol-level negotiation, and third-party library usage.

3) *G3: Safety-Security Co-Design for ICS PQC Migration:* As noted in Section V-F, no paper in our corpus addresses the formal safety-security co-design challenge for ICS/OT PQC migration. IEC 62443 [99] (industrial cybersecurity) and IEC 61508 (functional safety) have no intersection addressing PQC transition, a gap with potential for physical harm if cryptographic failures occur during migration.

4) *G4: Standardized Crypto-Agility Metrics:* The concept of crypto-agility [41] lacks quantitative metrics: no paper proposes a measurable crypto-agility score comparable to our MRC

Fig. 7 — Research Gap Heatmap
(Red = zero coverage; G1–G6 annotate critical gaps)

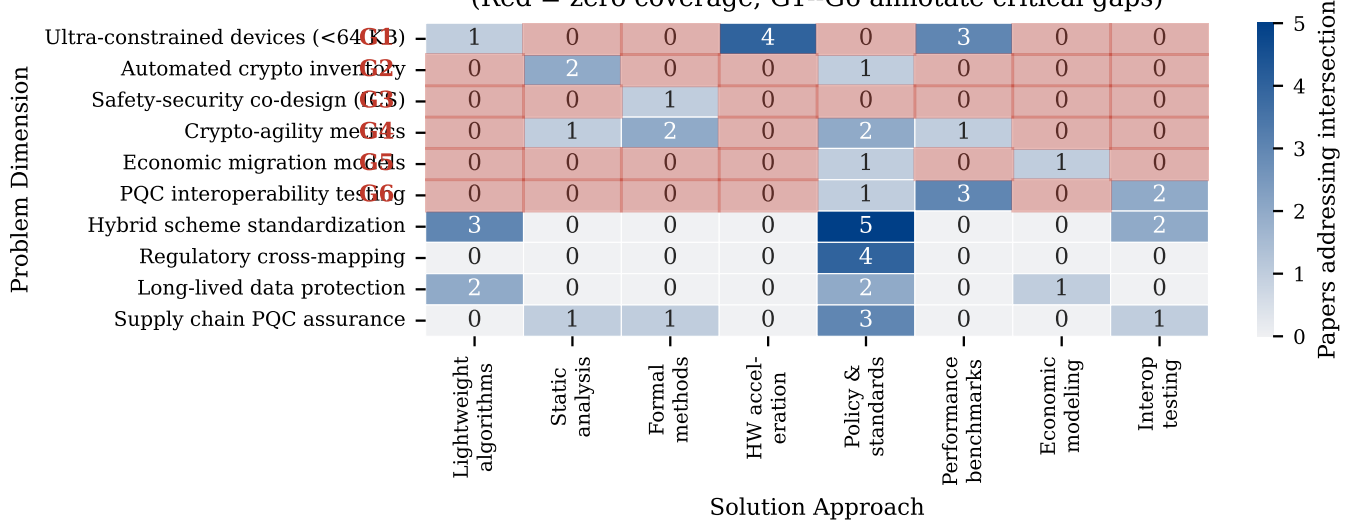


Fig. 11: Research gap heatmap: problem dimensions (rows) vs. proposed solution approaches (columns). Cell color intensity indicates number of papers addressing the intersection. Red cells indicate zero coverage (research gaps). White numbers indicate paper counts where ≥ 3 . Six major gaps (G1–G6) are annotated.

framework. This prevents organizations from benchmarking their agility posture or setting verifiable targets.

Ott et al. [116] survey the cryptographic transition literature and identify a research void in process-level migration tooling: the research community has produced strong algorithm implementations but almost no tooling for managing the *organizational* transition (dependency discovery, rollout orchestration, fallback procedures). Von Nethen et al. [117] address this gap partially with PMMP (Post-Quantum Cryptography Migration Management Process), a structured process model with 47 discrete migration tasks organized across five phases (Inventory, Assessment, Planning, Execution, Verification). PMMP provides a process scaffold but does not operationalize measurement. Gupta and Mittal [118] propose a cloud-specific readiness framework for enterprise SaaS that includes 12 measurable readiness indicators, the closest approximation to a quantitative crypto-agility metric in the post-standardization literature, though no consensus standard has emerged.

5) *G5: Economic Models for PQC Migration Cost*: Total cost of ownership (TCO) for PQC migration encompasses hardware replacement, software reengineering, certificate re-issuance, operational disruption, and re-testing. Few papers in our corpus address this dimension quantitatively. Campbell [119] provides the most detailed timeline analysis to date, projecting that Fortune 500 enterprises face 7–12 year full migration timelines with peak annual expenditure exceeding \$200M for organizations with large PKI deployments, driven by certificate lifecycle management costs rather than algorithm performance. Lohmiller et al. [120] quantify automotive migration costs specifically, identifying that ML-KEM/ML-DSA integration into vehicle ECUs requires hardware security module (HSM) upgrades at an estimated \$15–\$40 per vehicle at production scale, creating a \$3–\$8B industry-wide bill for

current global vehicle fleet refresh cycles. Sowa et al. [121] measure real-world PQC adoption rates across enterprise networks, finding that organizations that began migration planning before 2023 (triggered by NIST Round 3 announcements) demonstrate $4.2\times$ faster deployment rates than those beginning post-FIPS publication, suggesting that economic costs are highly sensitive to migration lead time. The $\approx \$1.5T$ estimated for Y2K remediation in 1999 provides a historical upper bound comparison; domain-specific cost modeling remains an open gap.

6) *G6: Interoperability Between PQC Implementations*: Algorithm conformance testing exists (NIST CAVP), but interoperability testing between independent implementations of the same PQC algorithm is absent as a systematic program. Incompatible polynomial encoding, byte ordering differences, and parameter set ambiguity have already caused interoperability failures in early ML-KEM deployments [63]. The absence of a standardized interoperability testing framework (analogous to TLS interop testing at IETF hackathons) creates systemic risk in multi-vendor PQC deployments.

7) *G7: Empirical Validation of Heuristic Migration Readiness Frameworks*: The MRC framework proposed in this survey (Section VI) assigns weights to the three readiness dimensions (Exposure, Feasibility, Data Sensitivity) using expert judgment. No published work in our corpus applies formal multi-criteria decision analysis methods—such as Analytic Hierarchy Process (AHP) or Delphi-based consensus elicitation—to validate or calibrate heuristic PQC migration scoring rubrics. A two-round Delphi survey with 14–19 PQC practitioners and security architects, followed by AHP consistency verification, is identified as the next immediate step for the MRC framework and as a methodological gap for the field more broadly. Until such validation is available, any heuristic readiness score should be

treated as a decision-support estimate rather than an empirically calibrated measurement.

X. FUTURE RESEARCH DIRECTIONS

A. AI/ML Augmentation of Migration Governance

The AI/ML techniques surveyed in Section VIII provide a foundation for continuous migration governance. Key open problems include: (a) standardized CBOM formats enabling tool interoperability across SCA and ML-based inventory tools; (b) adversarially robust classifiers for cryptographic binary detection; (c) RL-based rollout agents validated on real enterprise topologies beyond synthetic simulations; and (d) on-device inference for anomaly detection in resource-constrained ICS/IoT environments where offloading to cloud is operationally infeasible. Benchmark datasets analogous to MLPerf, covering side-channel detection and cryptographic binary classification, would accelerate progress and enable reproducible comparison across methods.

B. PQC in 5G/6G Networks

5G New Radio security relies on RSA-based certificate infrastructure for network function authentication [122]. 6G standardization (IMT-2030, expected 2030) should be PQC-native from the outset. The 3GPP SA3 working group has initiated study items on PQC integration, but no standardized migration path for deployed 5G networks exists [123].

C. Post-Quantum Zero-Knowledge Proofs

zk-SNARKs (used in privacy-preserving blockchain) rely on elliptic-curve bilinear pairings broken by quantum adversaries. Post-quantum ZKP systems based on hash functions (STARKs) or lattice commitments are active research areas [124], with deployment implications for privacy technologies and anonymous credentials.

D. Quantum Key Distribution: Appropriate Scope

QKD provides information-theoretic security but requires dedicated fiber infrastructure, limits key distribution distance to ≈ 100 km without repeaters, and costs $\approx \$50,000$ – $\$200,000$ per endpoint [51]. Our analysis suggests QKD is appropriate only for point-to-point links between high-value government or financial nodes with perpetual secrecy requirements. Enterprise-scale deployment remains economically and technically infeasible for the foreseeable future.

E. PQC Hardware: ASICs and FPGAs

Dedicated silicon for PQC (NTT accelerators, Gaussian sampling engines, hash co-processors) will drive adoption in constrained domains. Recent work on ASIC implementations of ML-KEM achieves < 0.1 ms per operation at < 50 mW [78]. Supply chain integration timelines for PQC silicon suggest 3–5 years from standard finalization to broad availability, creating a hardware gap for systems requiring upgrade before 2028.

XI. CONCLUSION

This survey has provided a structured narrative analysis of post-quantum cryptography migration across six critical infrastructure domains, synthesizing 148 peer-reviewed publications and 39 normative documents. To our knowledge, it is among the first surveys to address all six domains under a unified framework with explicit migration-readiness scoring. Our principal contributions are: (1) a three-level unified taxonomy enabling cross-domain comparison of PQC migration approaches; (2) the Migration Readiness Classification (MRC) framework, a heuristic decision-support rubric that scores domains using computed crypto-exposure, feasibility, and deadline dimensions, with PKI (0.788, T1) and VPN/ZTNA (0.888, T1) emerging as the most urgent; (3) identification of research gaps (particularly the absence of PQC solutions for ultra-constrained devices and the lack of safety-security co-design frameworks for ICS/OT migration) within the screened corpus; and (4) a cross-regulatory analysis revealing deadline misalignments across NIST, NSA CNSA 2.0, ENISA, and national frameworks with direct compliance implications for multinational operators.

The threat is not purely theoretical: intelligence assessments cite HNDL-style collection as a credible adversarial tactic [34], and data encrypted today with RSA/ECC may become vulnerable once a cryptographically relevant quantum computer emerges. The window for proactive migration is open but narrowing. Practitioners should initiate cryptographic inventories immediately, prioritize hybrid migration for TLS and PKI infrastructure, and engage with domain-specific regulators to establish sector compliance timelines. Researchers should urgently address gaps G1 (ultra-constrained PQC), G3 (ICS safety-security co-design), and G5 (economic migration models). The reproducible analysis code released alongside this survey provides a foundation for community-driven extension of the MRC framework and gap heatmap as the literature matures.

The era of quantum-safe cryptography is not a future concern. It is the engineering mandate of the present.

AI USAGE DISCLOSURE

The authors acknowledge the use of large language model (LLM) assistance for grammar refinement, bibliography formatting, and manuscript preparation support in the preparation of this manuscript, in accordance with IEEE's Authorship and AI Policy. All scientific content, survey design, methodology choices, data analysis, and conclusions are the sole responsibility of the human authors. Benchmark data are derived from publicly available open-source implementations (OQS liboqs [63] and pqm4 [45]).

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